

$$\bar{R}_i = \frac{15\% - 5\% + 10\% + 15\% + 3\%}{5} = 7.60\%.$$

3. Compute the fund's geometric mean annual return. How does it compare with the arithmetic mean annual return?

Solution:

The geometric mean annual return can be computed as:

$$\bar{R}_{Gi} = \sqrt[5]{1.15 \times 0.95 \times 1.10 \times 1.15 \times 1.03} - 1,$$

$$\bar{R}_{Gi} = \sqrt[5]{1.4235} - 1 = 0.0732 = 7.32\%.$$

Thus, the geometric mean annual return is 7.32 percent, which is slightly less than the arithmetic mean return of 7.60 percent.

4. The Lohrmanns want to earn a minimum annual return of 5 percent. The annual returns and investment amounts are presented in Exhibit 13. Is the money-weighted annual return greater than 5 percent?

Solution:

To calculate the money-weighted rate of return, tabulate the annual returns and investment amounts to determine the cash flows, as shown in Exhibit 13:

$$CF_0 = -30.00, CF_1 = -10.50, CF_2 = +22.75, CF_3 = -3.00, CF_4 = -6.25, CF_5 = +36.05.$$

We can use the given 5 percent return to see whether or not the present value of the net cash flows is positive. If it is positive, then the money-weighted rate of return is greater than 5 percent, because a 5 percent discount rate could not reduce the present value to zero.

$$\frac{-30.00}{(1.05)^0} + \frac{-10.50}{(1.05)^1} + \frac{22.75}{(1.05)^2} + \frac{-3.00}{(1.05)^3} + \frac{-6.25}{(1.05)^4} + \frac{36.05}{(1.05)^5} = 1.1471.$$

Because the value is positive, the money-weighted rate of return is greater than 5 percent. The exact money-weighted rate of return (found by setting the above equation equal to zero) is 5.86 percent.

These calculations can be performed using Excel using the =IRR(cash flows) function, which results in an IRR of 5.86 percent.

Time-Weighted Returns

An investment measure that is not sensitive to the additions and withdrawals of funds is the time-weighted rate of return. The **time-weighted rate of return** measures the compound rate of growth of USD1 initially invested in the portfolio over a stated measurement period. For the evaluation of portfolios of publicly traded securities, the time-weighted rate of return is the preferred performance measure as it neutralizes the effect of cash withdrawals or additions to the portfolio, which are generally outside of the control of the portfolio manager.